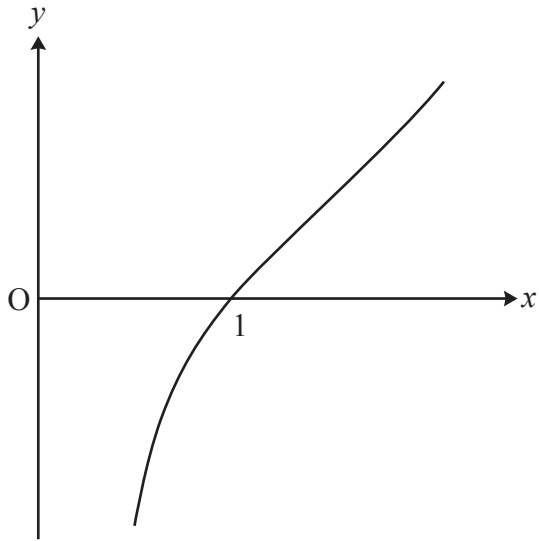
7 The diagram shows part of a curve which passes through the point (1, 0).



The gradient of the curve is given by $\frac{dy}{dx} = 6x + \frac{8}{x^3}$.

Determine whether the curve passes through the point (2, 12). **[6]**